Code Workflow Analysis: cancer.py

Breast Cancer Classification using Logistic Regression

Overview

The provided script, `cancer.py`, implements a machine learning pipeline to classify breast cancer tumors as either malignant or benign. It utilizes the Scikit-learn library for data handling and modeling, and Seaborn/Matplotlib for visualization.

Step-by-Step Workflow

1. Environment Setup & Dependencies

The script begins by importing necessary modules:

- `sklearn.datasets`: To fetch the built-in breast cancer dataset.
- `linear_model.LogisticRegression`: The classification algorithm.
- `model_selection.train_test_split`: To partition the data.
- `metrics`: To evaluate the model's performance.
- `seaborn/matplotlib`: To create visual representations of results.

2. Data Loading

The `load_breast_cancer()` function retrieves a dataset containing 30 numeric features (like radius, texture, and perimeter) and a target variable (0 for malignant, 1 for benign).

```
data = load_breast_cancer()
X = data.data   # Features
y = data.target   # Labels
```

3. Data Splitting

To ensure the model can generalize to unseen data, the script splits the dataset into two sets: **80% for training** and **20% for testing**. The `random_state=42` ensures the split is reproducible.

4. Model Initialization & Training

A Logistic Regression model is instantiated with `max_iter=5000` to give the solver enough iterations to converge. The `fit` method adjusts the model parameters based on the training features (`X_train`) and labels (`y_train`).

```
model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)
```

5. Prediction and Evaluation

After training, the model predicts labels for the test set. Two primary metrics are used to judge success:

- **Accuracy Score:** The percentage of total correct predictions.
- **Confusion Matrix:** A table showing True Positives, True Negatives, False Positives, and False Negatives.

6. Visualization

Finally, the `seaborn.heatmap` function converts the numeric confusion matrix into a graphical format, making it easier to identify where the model is making errors.

Conclusion

This script represents a standard supervised learning workflow. By using Logistic Regression, it attempts to find a linear boundary that best separates the two classes of cancer based on the provided diagnostic measurements.